

consumer world. Wearable technologies are starting to play a crucial role in the military as well. Heart rates, perspiration rates and several other vital signs are getting monitored for soldiers and communicated back via a data link, to monitor health and send required medical help during an emergency.

Also, the rapid development and deployment of high-speed wireless technologies like 3G and now LTE have enabled literally the whole world to be connected. Defence industry is now using a modified version of 3G, LTE and Wi-Fi protocols along with necessary security for military communication, instead of relying on proprietary protocols they had used in the past.

Q. Like wearables, is there anything happening in the field of the Internet of Things (IoT) on the military application side?

A. When we talk about the IoT in a military application, it is mostly around wireless sensor networks and related software protocols. However, everything starts from the sensor. Sensors, the signal chain that makes sense of data and converts it to digital, digital-signal processors, RF devices and Big Data are the main components of any IoT system. Our focus is on how to optimise the IoT in military via above-mentioned applications.

Another aspect of the IoT is energy harvesting, where we are working on several technologies on how best to

leverage sunlight, ambient light or vibration to energise sensors and devices in the field.

Q. Will energy harvesting be the primary area of research for attempts at reducing power consumption?

A. One of the advances that we see in this area is the need for extremely low power to power up these sensors and associated electronics, especially with the emergence of the IoT. While several new advances are being made in terms of how to reduce power consumption for sensors, the signal chain, the processor and wireless sensor networks, energy harvesting is the trend that is attracting major research funds. The idea is to see if energy could be harvested in the field to power up these remote sensors and edge-nodes of the IoT.

Q. What kinds of technologies are currently available for harvesting energy?

A. Wherever we put an integrated chip between a hot and a cold area, we can look at harvesting energy from a temperature difference standpoint. Energy can also be harvested from vibrations and movements. Ambient light sensors are another area to look at. Energy transmission and storage plays a key role here. So our focus is on building energy transmission and harvesting devices. ●

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